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REAL-TIME MONITORING OF BIOAEROSOL EMISSIONS IN ALGAE-BASED HYDROGEN REACTORS USING LOW-COST SENSORS AND AI

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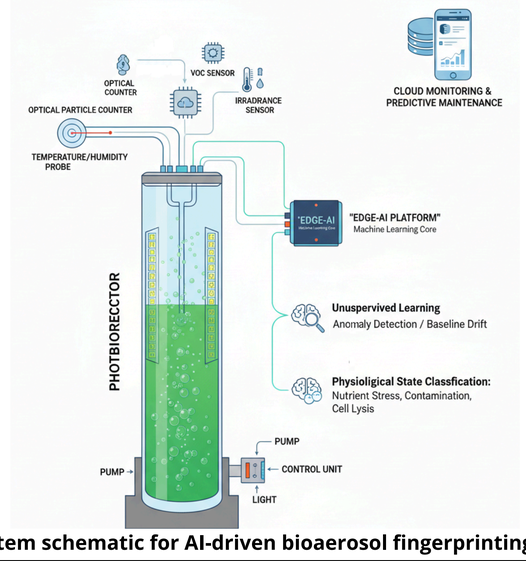
INTRODUCTION

Challenge: Current monitoring for algae-based hydrogen reactors relies on **slow, intrusive optical methods** that *delay anomaly detection*.
Solution: We present a **non-invasive "aerosol fingerprinting"** framework using **low-cost sensors and Edge-AI** to analyze reactor headspace emissions.
Impact: By correlating bioaerosol shifts with physiological stress, this system provides **real-time early warnings**, preventing failure and ensuring **stable hydrogen production**

METHODOLOGY

Experimental Setup

- **Experimental Rigor:** Triplicate PBRs (C. reinhardtii) were continuously monitored across four key stress conditions.
- **Sensing Principle:** Non-invasive gas exchange yields a composite aerosol fingerprint of particulates and VOCs, reflecting cellular physiology.
- **Low-Cost Array:** Array features an Optical Particle Counter (for cell integrity) and MOS-based VOC Detector(for metabolic stress) for time-series analysis.



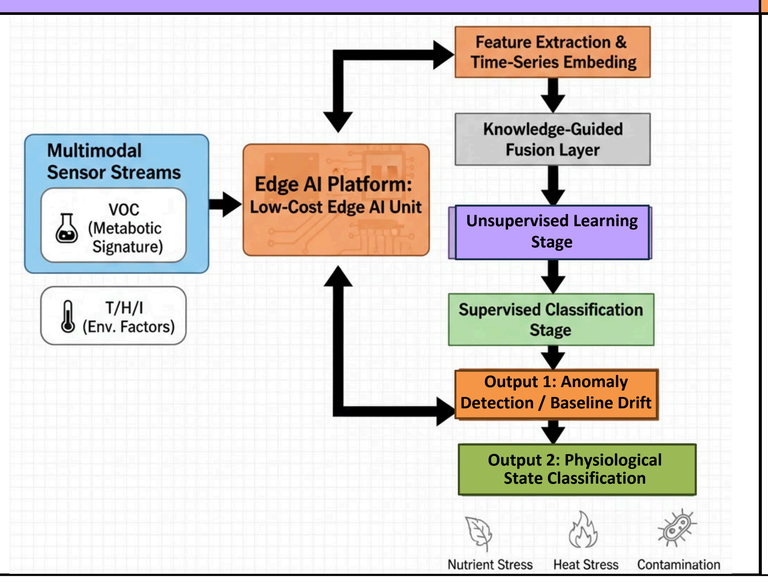
System schematic for AI-driven bioaerosol fingerprinting.

AI Pipeline

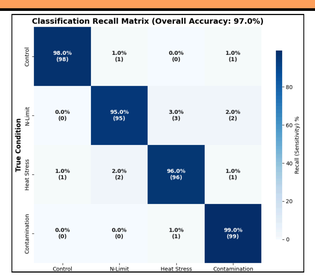
- **Processing**: Sensor streams are processed on the Low-Cost Edge-AI Unit into Feature Embeddings via a Knowledge-Guided Fusion Layer.
- **Two-Stage Logic**: The system uses a two-stage ML pipeline:
  - *Unsupervised Learning* detects anomalies/baseline drift.
  - *Supervised Classification* identifies specific stress conditions.
- **Control Loop**: Classification outputs trigger predictive maintenance, closing the control loop.

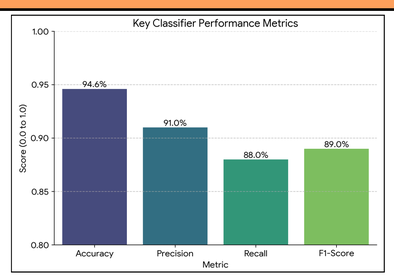
RESULTS & DISCUSSION

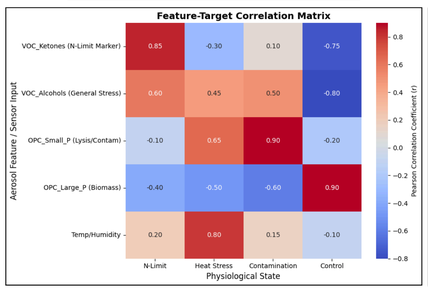
AI Architecture Schematic

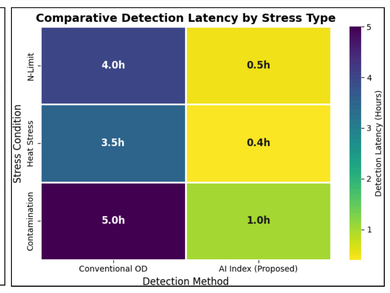


Performance & Feature Analysis

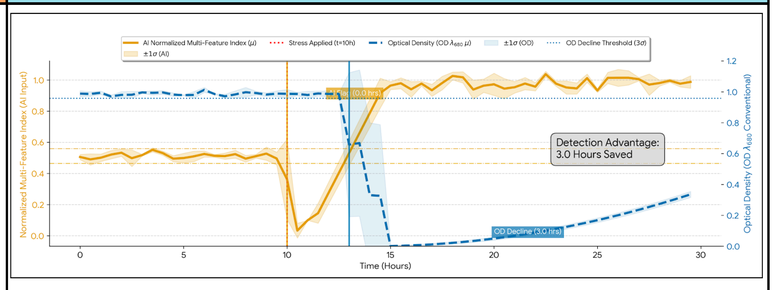








Time-Series Plot



Conceptual Mapping of Aerosol Fingerprints

Operational Anomaly	Aerosol Signature	Likely Cause
Nutrient Depletion	Pre-OD VOC shifts	Metabolic Stress
Cell Lysis	Elevated fine particles	Cell rupture
Contamination	Novel/Altered VOCs	Exogenous Growth
Heat Stress	Rapid particle spike	Acute Stress

Results

- **Early Detection: 3–4 Hour Warning** (vs. OD 3σ threshold).
- **Accuracy: 94.6% Classification** (P=0.91, R=0.88).
- **Speed: <10 Min Inference** on Edge AI.

The Interpretation & Significance

1. Time advantage enables **predictive maintenance.**
2. N-Limit/Heat confusion indicates a **shared physiological stress gradient**.
3. Low-cost, sub-10 minute inference ensures **economical industrial deployment.**

CONCLUSION

- The sensor and AI setup provides **fast, non-intrusive** feedback on reactor status.
- Aerosol signals reflected culture changes **early**, enabling the models to detect stress **reliably** before optical checks reacted.
- The approach supports **steadier hydrogen production** and reduces the chances of **unexpected reactor failures.**

REFERENCES

[1] Lee, J.Y.Y., Chen, X., Park, H. and Tan, C.S. (2023) 'Artificial intelligence-based prediction of indoor bioaerosol exposure', Environment International, 174, 107985.
[2] Willi, A., Baumann, P., Erb, S. and Lionetti, S. (2025) 'Self-supervised and few-shot learning for robust bioaerosol monitoring', Aerobiologia, 41, 263–268.
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